

Session Objectives

At the conclusion of the session/course, the student should be able to:

- Describe the presentation and diagnostic tests used to diagnose Mixed Connective Tissue Disease (MCTD)
- Describe the presentation and diagnostic criteria used to diagnose dermatomyositis & polymyositis.
- Discuss the presentations of Vasculitis.
- Discuss the overlap of presentation with non-rheumatologic diseases.

Presentation and Diagnostic Challenges

- Non-specific presentation
 - Fatigue
 - Myalgias
 - Arthralgias
 - Low-grade fevers
- Typically, will be diagnosed with one connective tissue disease initially and then other disorders present later.
 - *Raynaud phenomenon common*



Source: Mosca M, Tani C, Neri C, Baldini C, Bombardieri S. Undifferentiated connective tissue diseases (UCTD). *Autoimmun Rev.* 2006 Nov;6(1):1-4. Epub 2006 Apr 19.
Review. PubMed PMID: 17110308.

Differential Diagnosis of MCTD

- Rheumatoid arthritis
- SLE
- Scleroderma
- Sjögren syndrome
- Polymyositis/Dermatomyositis
- Eosinophilic fasciitis
- Graft-versus-host disease



Dermatomyositis and Polymyositis



Sex:
2:1 F:M

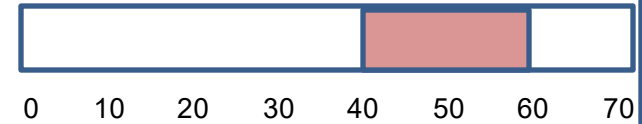
High rate of malignancy with DM:

- Ovarian
- Lung
- Pancreas
- Stomach
- Colon

Dermatomyositis

Path:

Cd4+ T lymphocyte
Damage perimysium
and blood vessels
Perifascicular
atrophy/necrosis



Peak Incidence 40-60 y/o

Peak Incidence 5-14 y/o

Polymyositis:

- Damages endomysium
- **Proximal b/l muscle weakness**
- Leg weakness occurs before UE
- No facial effects
- Painful
- Difficulty with swallowing

Dermatomyositis:

- Dark red malar rash
- Erythema on face, shoulders upper chest and back
- Periorbital edema:
- **heliotrope rash**
- **Gotttron papules**
- Nail-fold capillary dilation

Dermatomyositis and Polymyositis

Dermatomyositis:
Gottron Papules
Heliotrope rash
Nailfold Capillary
Dilation



Elevated CK
Muscle Biopsy
Special Tests:

- Anti-Jo-1 Ab most common
- Anti-Mi-2 Ab → DM
- Anti-SRP Ab → PM
- Anti-115/140 relates to risk of malignancy

Treatment

- Corticosteroids
- Methotrexate
- IVIG
- Rituximab
- Limit sun exposure

“Antisynthetase Syndrome

- Patient with findings of both PM and DM with findings similar to SLE arthritis

Myositis specific antibodies

Antibody	Clinical Association
Anti-Jo-1 and other antisynthetase antibodies	Polymyositis or dermatomyositis with interstitial lung disease, arthritis, mechanic's hands
Anti-Mi-2	Dermatomyositis with rash more than myositis
Anti-MDA5 (anti-CADM 140)	Dermatomyositis with rapidly progressive lung disease, cutaneous ulcers
Anti-155/140	Cancer-associated myositis
Anti-140	Juvenile dermatomyositis
Anti-SAE	Cancer-associated dermatomyositis, dermatomyositis with rapidly progressive lung disease
Anti-signal recognition particle	Severe, acute necrotizing myopathy
Anti-HMG CoA reductase	Necrotizing myopathy related to statin use

Vasculitis Pathology

- Immune complex deposition- walls of vessels
 - Activate complement > neutrophils > lysosomal enzymes cause damage.
 - Examples: IgA vasculitis, Polyarteritis nodosa.
- Antineutrophil cytoplasmic antibodies (ANCA)
 - Target components of neutrophils, monocyte lysosomes, and endothelial cells
 - GPA and MPA
- Anti-endothelial cell antibodies
 - GPA, Kawasaki, Takayasu
- Autoreactive T cells.
 - Cytokine and macrophage activation

What

Vasculitis

- When to consider:
 - Systemic illness + ischemia otherwise unexplained
 - Elevated **ESR** and CRP
 - Biopsy
 - Combinations of renal, lung, and skin diseases (petechiae/purpura).

Large Vessel

Giant Cell Arteritis

- Granulomatous T-cell mediated
- Aorta and/or carotid, vertebral, and temporal arteries
- Age >50
- Symptoms: *headache, diplopia, vision loss, fever / weight loss / fatigue*
- Dx: Biopsy

Takayasu Arteritis

- Granulomatous T-cell mediated
- Age <50
- Symptoms: *ocular disturbances, weakening of upper ext. pulses.*
- Fibrous thickening of the aorta

Medium Vessel

Thromboangiitis obliterans Buerger Disease

- Associated with thrombosis and distal ischemia in extremities.
- Young males <35, heavy smokers
- Raynaud phenomenon, extreme pain, to gangrene

Polyarteritis nodosa

- Necrotizing
- Doesn't affect lung
- Affects: skin, kidney > heart > liver > gi tract
- Not assoc. with ANCA's
- 1/3 have hepatitis B.

Medium Vessel

Kawasaki disease

- Mucocutaneous lymph node syndrome
- Coronary arteries: aneurysm, thrombosis, rupture
- Infants and young children 80% <5 yo
- Acute, febrile, self-limited
- Delayed type hypersensitivity after viral infection.

Small Vessel

ANCA associated:

- Microscopic polyangiitis
 - Lungs and kidney
- Granulomatosis with polyangiitis “Wegener granulomatosis”
 - Nasopharynx, lungs, kidneys
- Eosinophilic granulomatosis with polyangiitis “Churg-Strauss syndrome”
 - Pulmonary, also heart and CNS

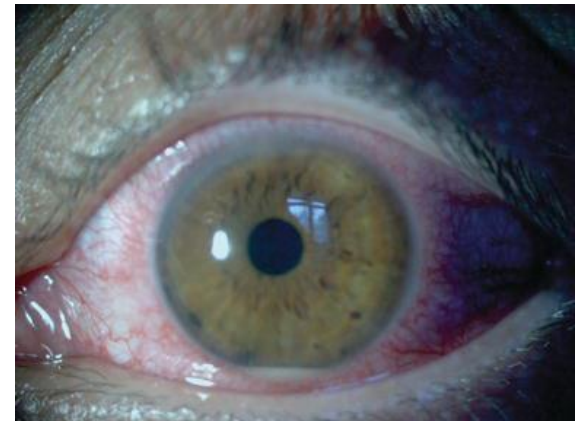
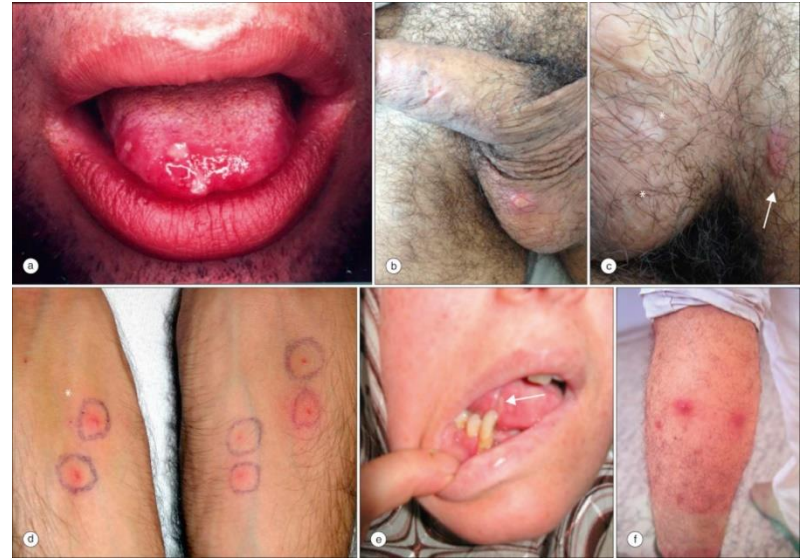
- Immune complex vasculitis
 - Deposits of immunoglobulin and or complement. Glomerulonephritis is frequent.
 - IgA vasculitis. “Henoch-Schonlein purpura”
 - children

Presentation of Behcet's Disease

Variable vessel vasculitis

Classic presentation is a patient presenting with reoccurring:

- **Oral** and genital ulcers
- Uveitis
- Joint and muscle pain
- Also affects: GI tract, CNS, and cardiovascular system



Source:

Behçet syndrome. Elsevier Point of Care Updated December 17, 2019.

Treatment of Behcet's Disease

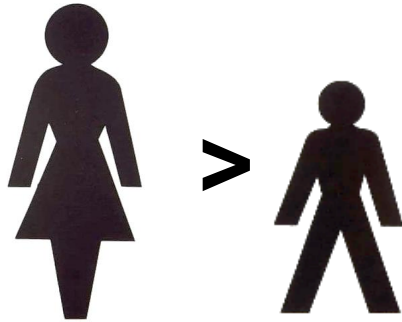
- Treatment is scaled depending on symptom presentations
 - Corticosteroids (topical/oral/IV)
 - Colchicine
 - Immunosuppressants
 - TNF-alpha antagonists
- Prognosis: Males worse than females, no lifetime remission but some patients have decreased symptoms with age

Differentials:

Myopathy vs. Weakness

- Infectious vs. non-infectious
- Medications
 - Example: HMG CoA reductase inhibitor
 - (statins)
- Toxins/Alcohol
- Neurologic Pain
 - Fibromyalgia
- Congenital
- Endocrine disorders:
 - Example hypothyroidism

Fibromyalgia Considerations



Sex:

More commonly
diagnosed in Women
than Men



0 10 20 30 40 50 60 70 80

Can occur at any age but typical
onset late teens to early 40s



Disease of
exclusion – Not
auto-immune

Associated with chronic conditions,
including rheumatologic diseases.
Musculoskeletal pain
No association with autoantibodies